

Atomic Layer Deposition of $\text{Li}_x\text{Al}_y\text{S}$ Solid-State Electrolytes for Stabilizing Lithium-Metal Anodes

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Increasing global demands for energy storage in electric vehicles and the grid necessitate “beyond-lithium-ion” batteries that simultaneously exhibit high capacity, long-term cyclability, and excellent safety. Several candidates, most notably lithium–air and lithium–sulfur batteries, utilize lithium-metal anodes. Although the specific capacity of lithium metal is unrivaled (3860 mAh g^{-1}), the capacity fade and risk of catastrophic failure must be overcome. Herein, we describe ultrathin, ionically conducting lithium aluminum sulfide ($\text{Li}_x\text{Al}_y\text{S}$) coatings prepared on lithium-metal anodes by using atomic layer deposition, which stabilize the Li–electrolyte interface and reduce (up to five times) the interfacial impedance of lithium–metal anodes in contact with organic electrolyte. We demonstrate that these thin films effectively suppress dendrite formation and preserve a smooth, low-surface-area morphology for electrodeposited lithium metal, thereby doubling the lifetime of Li–Cu asymmetric cells.

■ ■ Please give academic titles (Prof./Dr.) for each author ■ ■

Renewable, green alternatives to fossil fuels are needed to address the twin challenges of satisfying the worldwide demand for electrical power and mitigating the threat of global warming. Among the alternatives, solar and wind power appear to be the most promising, but the intermittent and distributed nature of these energy sources demands radical advances in energy-storage technology to facilitate their large-scale deployment. Consequently, tremendous effort is being dedicated toward developing “beyond-lithium-ion” batteries that can meet the demands of transportation and grid storage. A number of candidate technologies, including Li–air and Li–sulfur batteries, hold great promise in this regard, but significant challenges must be overcome before these batteries can be implemented. In particular, the lithium-metal anodes used

in both Li–air and Li–sulfur batteries are a source of capacity fade and catastrophic failure.

Lithium metal has an extremely high theoretical specific capacity (3860 mAh g^{-1}), low density (0.59 g cm^{-3}), and the lowest negative electrochemical potential of any known material (-3.040 V vs. the standard hydrogen electrode). These attributes make lithium metal extremely attractive as an anode material for rechargeable lithium batteries. Unfortunately, although lithium metal has been investigated as an anode over the last 40 years,^[1] its commercialization has been hindered by two severe technical barriers: the growth of lithium dendrites during repeated charge/discharge processes and the low coulombic efficiency (CE) of these processes.^[2] The low CE can be partially compensated by using an excess of Li. However, the risk that lithium dendrites can grow to pierce the separator in a battery and reach the cathode, causing an electrical short and possibly fire or explosion, has compelled lithium-ion battery (LIB) manufacturers to adopt graphite anodes (372 mAh g^{-1})^[3] rather than lithium, thereby sacrificing capacity in favor of reliability and safety. Although existing LIBs are adequate for portable electronics, they are insufficient to enable the wide-scale implementation of electrical vehicles. Consequently, new battery technologies such as lithium–sulfur (Li-S)^[4] and lithium–air (Li-O_2)^[5] with up to four times higher volumetric energy density compared to LIBs, are undergoing intensive investigation. The success of these new technologies hinges on solving the longstanding challenges posed by lithium-metal anodes.

Various strategies have been explored to prevent lithium dendrite growth, but, to date, these have not proved commercializable. Some approaches have utilized micro-structured anodes such as coated lithium powder^[6] or a 3D porous current collector^[7] to suppress dendrite formation. However, most strategies have focused on improving the stability of the in situ formed solid electrolyte interphase (SEI) layer through electrolyte additives,^[8] by applying ex situ (i.e. artificial) SEI layers to the lithium metal prior to battery fabrication^[9] and by micro-structuring the anode.^[6,7] For instance, Archer and co-workers achieved stable, long-term cycling of Li by adopting simple liquid electrolytes reinforced with halogenated salt blends.^[8a] Zhang and co-workers reported that Cs^+ additives in the electrolyte can form a positive electrostatic shield around the initial nucleating points for lithium dendrites, which inhibit further growth.^[8b] Electrolytes with high salt concentration and large Li-ion transference numbers can also effectively suppress lithium dendrite growth.^[8c,d] Numerous materials have been investigated as artificial SEI layers, including polymers,^[9a,b] SiO_2 ,^[9c] Li_3N ,^[9d,e] and carbon materials.^[9f,g] For instance, Cui and co-workers utilized a monolayer of interconnected hollow carbon

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nanospheres to protect lithium metal and demonstrated that lithium was deposited underneath the carbon layer without forming dendrites.^[9g] In addition, monolayer hexagonal boron nitride and graphene can also act as protective layers for lithium metal.^[9h]

Ideally, artificial SEI layers on lithium metal should be uniform, conformal, ionically conducting, stable against liquid organic electrolytes, and deposited at low temperatures (lithium melts at 180 °C). To this end, atomic layer deposition (ALD) is a promising technique. ALD has attracted increasing attention in the development of next-generation LIBs in recent years, as a means to synthesize nanophase anodes, cathodes, and electrolytes.^[10] But, the greatest commercial impact for ALD in energy storage is likely through depositing ultrathin protective coatings. Numerous reports exist for ALD coatings on a variety of cathode and anode materials, but only very recently has ALD been investigated for protecting lithium-metal anodes and, thus far, the reports are limited to ALD of Al_2O_3 .^[11]

Herein, we report the use of ALD solid-state electrolyte thin films to protect lithium-metal anodes and inhibit the growth of Li dendrites. We developed a process for growing ALD lithium aluminum sulfide ($\text{Li}_x\text{Al}_y\text{S}$) films with precisely controlled thickness and composition and showing a relatively high ionic conductivity of $2.5 \times 10^{-7} \text{ S cm}^{-1}$ at room temperature. We demonstrate that this material can stabilize the interface between lithium metal and liquid carbonate electrolytes, and effectively suppress lithium dendrite growth.

The $\text{Li}_x\text{Al}_y\text{S}$ ALD was performed by combining an existing process for Li_2S ALD, using lithium tert-butoxide (LTB) and

H_2S ,^[12] with a new process for Al_2S_3 ALD using tris(dimethylamido) aluminum(III) (TDMA-Al) and H_2S . By controlling the ratio of Li_2S to Al_2S_3 ALD cycles, we can deposit $\text{Li}_x\text{Al}_y\text{S}$ films with any composition, as illustrated in Figure 1a. In this work, we focus exclusively on films using an ALD cycle ratio of $\text{Li}_2\text{S}/\text{Al}_2\text{S}_3 = 1:1$.

Figure 1b shows the mass changes observed through in situ quartz crystal microbalance (QCM) measurements during $\text{Li}_x\text{Al}_y\text{S}$ (1:1) ALD at 150 °C. The $\text{Li}_x\text{Al}_y\text{S}$ (1:1) growth is quite linear, and shows mass increases of approximately 13.1 ng cm^{-2} during each Li_2S cycle and 12.4 ng cm^{-2} during each Al_2S_3 cycle. Assuming bulk densities for these compounds, these mass changes imply thickness changes of 0.79 Å per cycle for Li_2S and 0.53 Å per cycle for Al_2S_3 , so that the average growth rate was 0.66 Å per cycle. Ex situ spectroscopic ellipsometry (SE) measurements performed on $\text{Li}_x\text{Al}_y\text{S}$ (1:1) deposited on Si(100) yielded a refractive index of $n = 2.77$ at 633 nm and a growth rate of 0.50 Å per cycle. This smaller growth compared to the value derived from QCM may result from interactions between the Li_2S and Al_2S_3 to produce a film that is denser than either of the pure compounds. The QCM measurements can also be used to estimate the composition of the $\text{Li}_x\text{Al}_y\text{S}$ (1:1) ALD films. Assuming stoichiometric growth of Li_2S and Al_2S_3 , the QCM measurements for $\text{Li}_x\text{Al}_y\text{S}$ (1:1) imply an atomic ratio of Li/Al of approximately 3.5. Ex situ measurements performed on $\text{Li}_x\text{Al}_y\text{S}$ (1:1) films deposited on Si(100) substrates by using inductively coupled plasma mass spectrometry (ICP-MS) yielded Li/Al = 2.9, which is in good agreement with the QCM measurements.

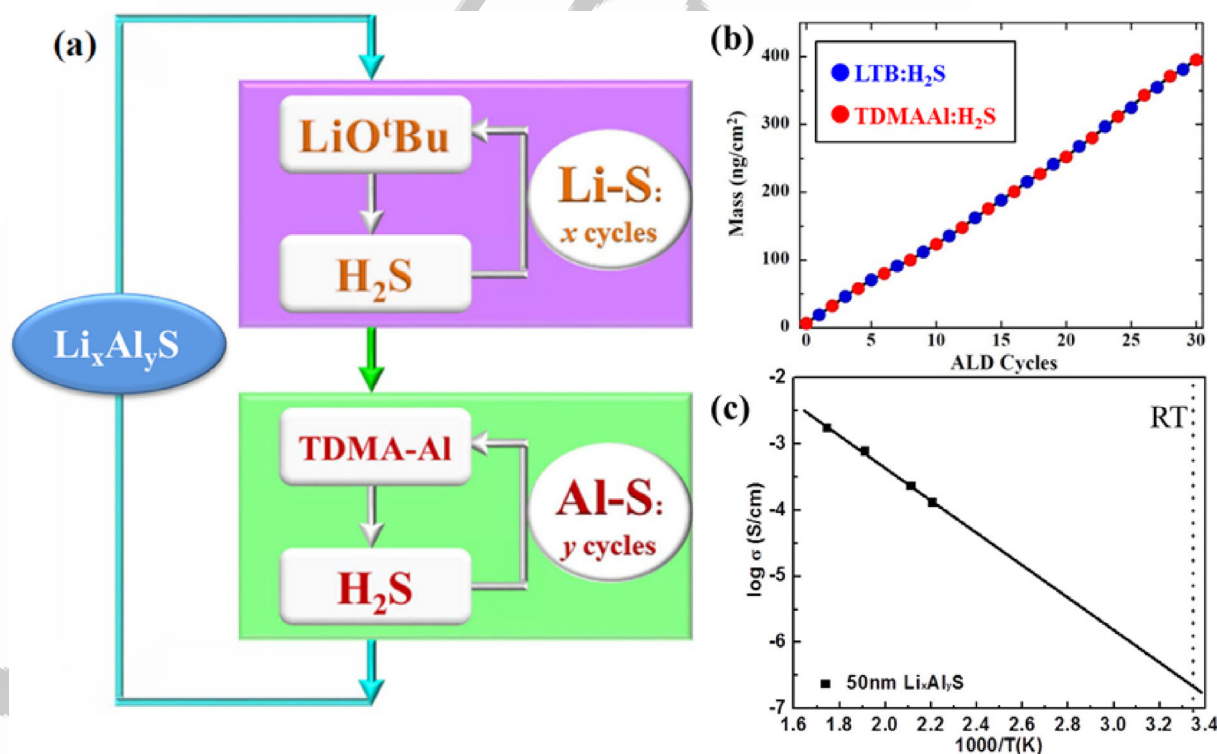


Figure 1. a) Schematic illustration of $\text{Li}_x\text{Al}_y\text{S}$ ALD; b) in situ QCM measurements of $\text{Li}_x\text{Al}_y\text{S}$ ALD performed at 150 °C using (LTB- H_2S)/(TDMAAl- H_2S) = 1:1; c) Arrhenius plot for temperature-dependent ionic conductivity measured using a 50 nm $\text{Li}_x\text{Al}_y\text{S}$ film.

To characterize the electrical properties of the $\text{Li}_x\text{Al}_y\text{S}$ (1:1) ALD film, we performed electrochemical impedance spectroscopy (EIS) measurements on 50 nm films deposited on SiO_2/Si substrates patterned with micrometer-scale interdigitated Pt electrodes in the temperature range of 180–300 °C. The resulting Nyquist plots (Figure 2) are composed of one semicircle and a tail. This finding indicates that our $\text{Li}_x\text{Al}_y\text{S}$ ALD films are amorphous, as crystalline materials would show two semicircles corresponding to separate contributions from grains and grain boundaries.^[13]

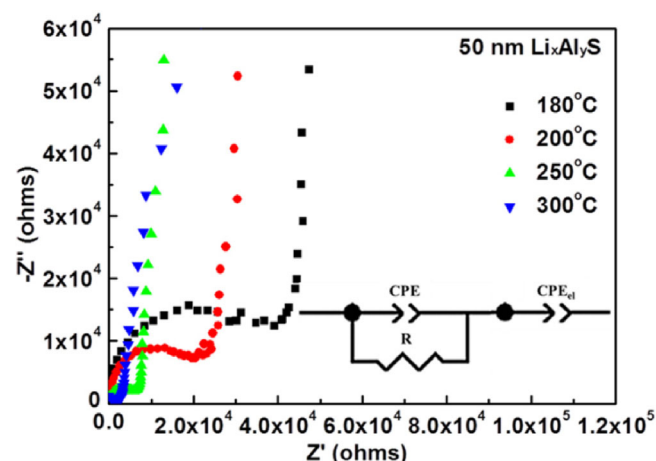


Figure 2. Nyquist plots of a 50 nm $\text{Li}_x\text{Al}_y\text{S}$ film. Insert shows the equivalent circuit.

The low-frequency tail is attributed to ionic blocking by the Pt electrodes. The ionic conductivity was calculated from the impedance data by fitting the equivalent circuit shown in the insert of Figure 2, where R is the resistance, CPE is the constant phase element, and the subscript “el” represents the electrode.^[14] The temperature dependence of the ionic conductivity follows Arrhenius behavior, as shown in Figure 1c, and yields an activation energy of approximately 0.48 eV. The extrapolated room-temperature ionic conductivity is $2.51 \times 10^{-7} \text{ S cm}^{-1}$ and is among the highest reported for ALD solid-state electrolytes.^[15] Moreover, the ionic conductivity could likely be improved by optimizing the composition of the $\text{Li}_x\text{Al}_y\text{S}$ films.

Next, we investigated the interfacial stability of lithium-metal anodes coated by the $\text{Li}_x\text{Al}_y\text{S}$ ALD film in contact with a commercial carbonate-based electrolyte (1 M LiPF_6 in EC/EMC 3:7) by fabricating Li/electrolyte/Li symmetric batteries and performing AC impedance measurements. As shown in the inset of Figure 3a, the impedance spectra are modeled by using an equivalent circuit composed of the electrolyte resistance R_1 , the resistance across the SEI R_2 with capacity CPE_1 , and the charge-transfer resistance at the electrodes R_3 with capacity CPE_2 .^[16] The AC impedance data in Figure 3a show only one semicircle, indicating that the time constants for CPE_1 and CPE_2 are nearly identical and cannot be resolved in these measurements. Figure 3b shows the R_{SEI} values for the symmetric batteries derived from the AC impedance measurements as a function of the storage time. After 68 h of storage,

the control sample with pristine Li showed a much higher R_{SEI} value of approximately 2400 ohms compared to that for the Li protected by the $\text{Li}_x\text{Al}_y\text{S}$ ALD film (ca. 600 ohms). Moreover, the R_{SEI} for pristine Li continued to increase throughout the test, whereas the coated Li stabilized after approximately 20 h. The continual increase in R_{SEI} for pristine Li indicates continual SEI formation from the reaction between the bare lithium and the electrolyte. Evidently, $\text{Li}_x\text{Al}_y\text{S}$ can protect the Li from direct contact with the electrolyte to produce a thinner and more stable SEI.

The CE is a defining parameter for determining battery lifetime. In our study using asymmetric Li/Cu cells, the CE is defined as the ratio of lithium stripped from the Cu electrode to lithium deposited on the Cu electrode during each cycle. Figure 3c shows the CE versus cycle number for asymmetric Li/Cu cells, using both pristine Cu electrodes and Cu coated with $\text{Li}_x\text{Al}_y\text{S}$ ALD films (1:1). The Li/pristine Cu cell shows a gradual decrease in CE over the first 90 cycles, followed by a precipitous drop at around 135 cycles. In comparison, the Li/ $\text{Li}_x\text{Al}_y\text{S}$ -coated Cu cell shows a very stable CE, extending until 170 cycles. The maximum CE values in Figure 3c are only around 80%, as a result of the carbonate-based electrolyte used.^[9f] A much higher CE of approximately 97% was achieved by using an ether-based electrolyte (Figure 4), but the behavior was very similar: the $\text{Li}_x\text{Al}_y\text{S}$ -protected Cu electrode maintained a high CE for over 700 cycles compared to only 400 cycles for the pristine Cu. In summary, the $\text{Li}_x\text{Al}_y\text{S}$ ALD film on the Cu electrode allows the Li/Cu cells to maintain a high and stable CE and long-term cyclability in both carbonate- and ether-based electrolytes.

In addition to the CE, the voltage hysteresis during Li deposition/stripping is another revealing parameter for long-term cyclability. We define hysteresis as the potential difference between Li stripping and Li deposition. For the Li/Cu cell with pristine Cu, the hysteresis increases with cycling, reaching 380 mV after 160 cycles (Figure 3d). We attribute this increase to the degradation and consumption of the electrolyte.^[9g, 16] For $\text{Li}_x\text{Al}_y\text{S}$ -protected Cu, however, the hysteresis is much more stable, remaining at approximately 120 mV over 120 cycles, and increasing only slightly to 157 mV after 160 cycles. These findings support the hypothesis that the $\text{Li}_x\text{Al}_y\text{S}$ ALD film inhibits any reaction between the electrodeposited Li and the electrolyte.

To investigate the origin of the improved electrochemical cycling performance imparted by the $\text{Li}_x\text{Al}_y\text{S}$ ALD film, scanning electron microscopy (SEM) was performed on the Li electrodeposited onto pristine Cu and $\text{Li}_x\text{Al}_y\text{S}$ -coated Cu (Figure 5). As expected, the Li deposited on pristine Cu shows numerous long filaments or nanowires^[9 g] (Figures 5a and 5c). It is easy to imagine that these 0.1–0.2 μm needle-like Li structures could easily puncture a separator in a Li battery, resulting in short circuit and failure. In addition, these structures dramatically increase the Li surface area compared to a planar foil, which would rapidly consume electrolyte to form an SEI. In contrast, the surface of the ALD $\text{Li}_x\text{Al}_y\text{S}$ -coated Cu remains relatively flat with no obvious Li dendrites after lithium deposition (Figure 5b). Figure 5d shows a cross-sectional SEM image of

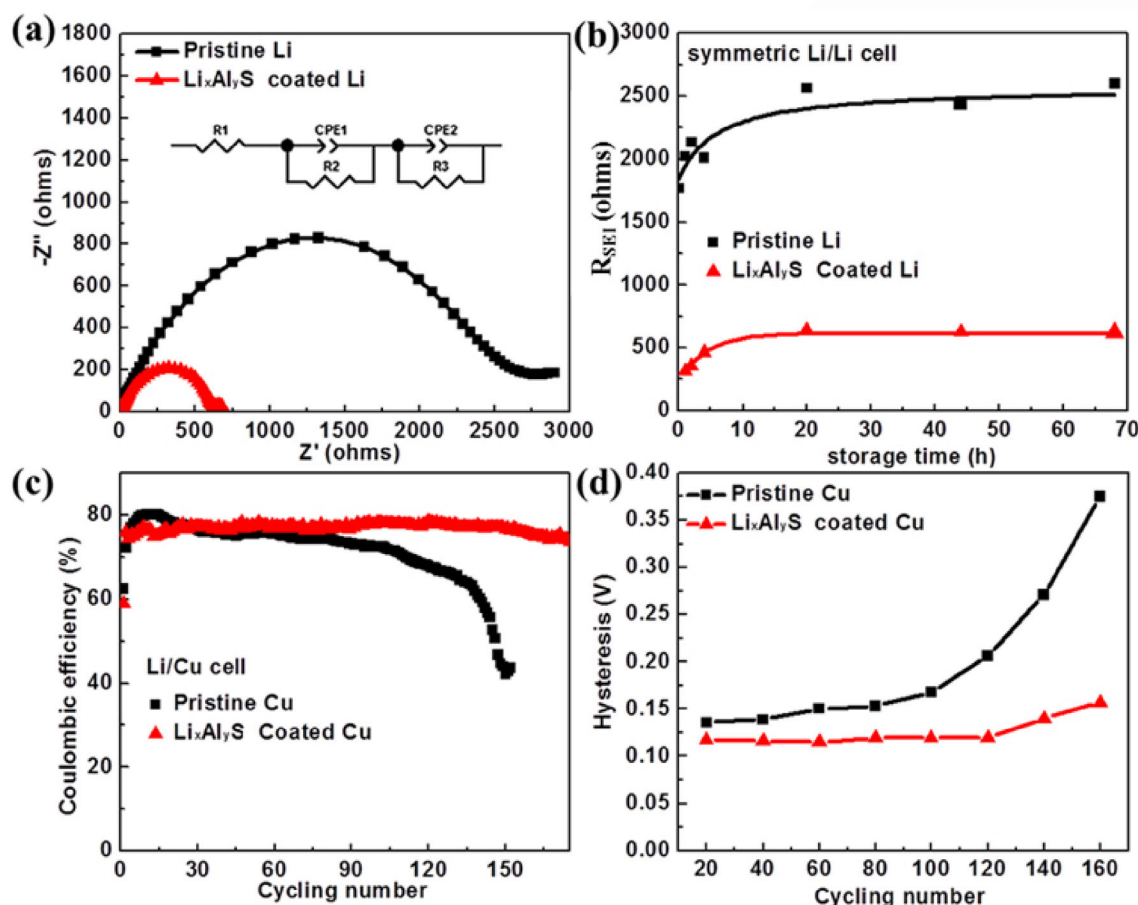


Figure 3. a) AC impedance spectra of Li/electrolyte/Li symmetric cells using both pristine Li and Li_xAl_yS-coated Li after 68 h storage; b) R_{SEI} of Li/electrolyte/Li symmetric cells versus storage time; c) CE of Li deposition/stripping in 1 M LiPF₆-EC/EMC with pristine Cu and Li_xAl_yS-coated Cu; d) comparison of hysteresis of Li deposition/stripping for both pristine Cu and Li_xAl_yS-coated Cu with Li metal as reference/counter electrodes.

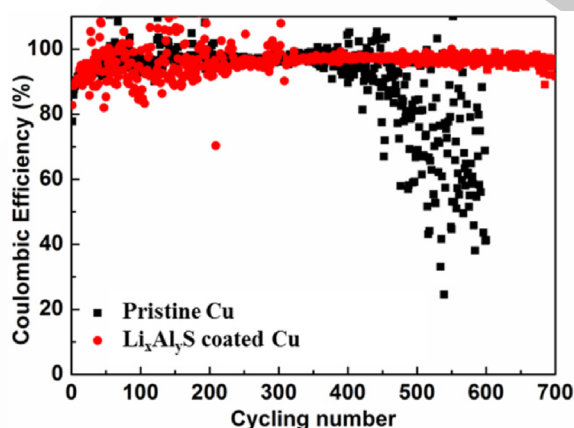


Figure 4. CE of Li deposition/stripping in 4 M LiTFSI-DME with pristine Cu or 25 nm Li_xAl_yS-coated Cu.

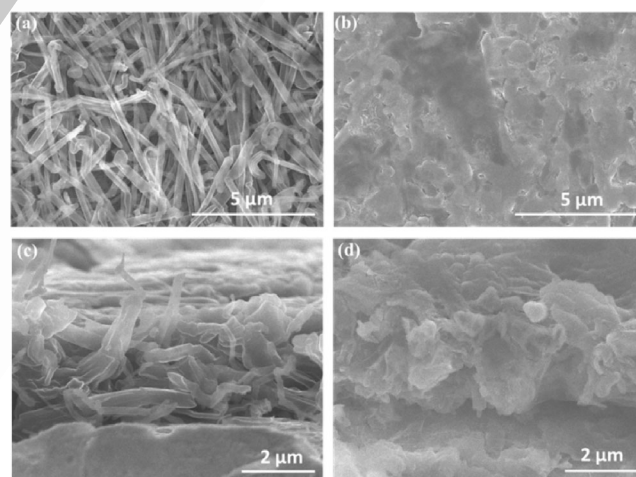


Figure 5. Top-view (a, b) and cross-sectional (c, d) SEM images of Li metal deposited on pristine Cu (a, c) and Cu coated with 50 nm Li_xAl_yS ALD film (b, d). Note that the 50 nm Li_xAl_yS ALD film is not visible in these images.

the ALD Li_xAl_yS-coated Cu after lithium deposition, and illustrates that the Li is much denser compared to the pristine Cu (Figure 5c). It can be concluded that the ALD-derived solid-state electrolyte Li_xAl_yS film can suppress Li dendrite growth.

In summary, we successfully synthesized Li_xAl_yS solid-state electrolyte thin films through ALD. These films show a high ionic conductivity of $2.51 \times 10^{-7} \text{ S cm}^{-1}$ at room temperature

and are highly effective in stabilizing the interface between lithium metal and liquid organic electrolytes. In addition, $\text{Li}_x\text{Al}_y\text{S}$ can effectively suppress lithium dendrite growth during Li deposition/stripping processes. These solid electrolyte ALD coatings show promise for improving the performance of Li-S and Li-O₂ batteries.

Experimental Section

We utilized a custom viscous-flow, hot-walled ALD reactor^[17] composed of a stainless-steel flow tube with a length of 100 cm and an inner diameter of 5 cm. Ultrahigh purity Ar (99.999%) was used as the carrier gas at a mass flow rate of 252 sccm and a pressure of 1.2 Torr. The Li_2S ALD^[12] was performed by using alternating 5 s exposures to lithium tert-butoxide (LTB, 98%, Strem Chemicals, Inc., USA) and hydrogen sulfide (1% H_2S in Ar, Matheson Tri-gas, USA), with 5 s Ar purging periods between each dose. The Al_2S_3 ALD was accomplished by using alternating exposures to 5 s tris(dimethylamido)aluminum(III) (TDMA-Al, Sigma-Aldrich Co., USA) and 10 s H_2S , separated by 5 s Ar purging periods. The solid LTB and TDMA-Al were heated to 140 and 80 °C, respectively, in stainless-steel reservoirs, and 36 sccm Ar was diverted through each reservoir during the respective precursor dose. The $\text{Li}_x\text{Al}_y\text{S}$ ALD was performed by alternating between the LTB- H_2S exposures for Li_2S ALD and the TDMA-Al- H_2S exposures for Al_2S_3 with every ALD cycle. In situ QCM measurements were performed to investigate the $\text{Li}_x\text{Al}_y\text{S}$ ALD. These experiments used RC-cut, $\blacksquare$ please define RC $\blacksquare$ 6 MHz quartz sensors (Colnatec, Inc.) mounted in a Maxtek BSH-150 sensor head. The sensor head was modified to purge the back side of the crystal with Ar to ensure that ALD only occurred on the front side.^[17] The QCM thickness monitor (Maxtek TM-400) reported mass changes in units of ng cm^{-2} , and these values were multiplied by the bulk densities of 1.66 and 2.32 g cm^{-3} for Li_2S and Al_2S_3 , respectively, to obtain the corresponding thickness changes.

To prepare $\text{Li}_x\text{Al}_y\text{S}$ ALD samples for testing, 50 nm films were deposited on Li and Cu foils at 150 °C. Both Li_2S and Al_2S_3 are extremely air-reactive. Consequently, the coated foils were unloaded from the ALD reactor into an Ar-purged glove bag, and placed in Ar-filled, sealed glass containers for transport into the glove box. The Li foils were loaded into the ALD reactor by using a similar protocol. Film thicknesses were determined from $\text{Li}_x\text{Al}_y\text{S}$ ALD films deposited on Si(100) substrates by using spectroscopic ellipsometry (J. A. Woollam Co. alpha-SE) performed in an Ar-purged glove bag.

To prepare samples for ionic conductivity measurements, 50 nm $\text{Li}_x\text{Al}_y\text{S}$ ALD films were deposited onto SiO_2/Si substrates patterned with micrometer-scale interdigitated Pt electrodes^[18] (Alta Microtec, Inc., Figure S3). These interdigitated Pt electrodes provide a 40000 $\times$ enhancement in ionic current compared to a simple arrangement of two electrodes in square geometry. In-plane ionic conductivity measurements were performed on the coated substrates in an Ar-filled glovebox by using EIS. Contact to the Pt electrodes was made by using microprobes, and the sample was placed on a hotplate to facilitate measurements in the temperature range of 180–300 °C. EIS was performed by applying 5 mV amplitude in a frequency range of 1 MHz–0.1 Hz, using a Solartron 1260 impedance/gain phase analyzer combined with a Solartron 1287 electrochemical interface.

CR2032-type coin cells were fabricated to facilitate impedance measurements and electrochemical cycling tests. Li/Li symmetric

cells were assembled with Li metal as both the working and counter electrodes. For the Li/Cu cells, the pristine Cu or $\text{Li}_x\text{Al}_y\text{S}$ -coated Cu foil was used as the working electrode, and Li foil was used as the counter and reference electrodes. Celgard 2400 membrane was used as the separator, whereas Cu foil served as substrates for the Li-metal deposition. The electrolyte used was 1 M LiPF_6 in EC/EMC (3:7). To standardize the testing, 30 μL of electrolyte was used in each coin cell. EIS of Li/Li cells was measured by using a Solartron 1260 impedance/gain phase analyzer combined with a Solartron 1287 electrochemical interface. Spectra were recorded from 1 MHz to 0.1 Hz with the amplitude of 5 mV. The spectra were fit by using the equivalent circuit in Figure 2a. For the electrochemical cycling testing of the Li/Cu cells, the current density for lithium deposition and stripping was set to 0.5 mA cm^{-2} . In each cycle, 1 C cm^{-2} (0.28 mAh cm^{-2}) of lithium metal was deposited on the Cu electrode and then stripped until the potential reached 1.0 V versus Li⁺/Li. The charge/discharge testing was performed by using an Arbin 2043 electrochemical tester.

The morphology of the electrodeposited lithium was characterized by using SEM (Hitachi S-4700). After a single half-cycle of lithium electrochemical deposition, the Li/Cu coin cells were disassembled to collect the lithium-metal film deposited on the Cu electrodes. Then, the electrolyte on the lithium surface was flushed away with DME. After drying, the Cu samples were transported to the scanning electron microscope in Ar-filled glass containers and quickly loaded into the microscope to minimize air exposure.

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Keywords: atomic layer deposition • dendrites • energy storage • lithium • solid-state electrolytes

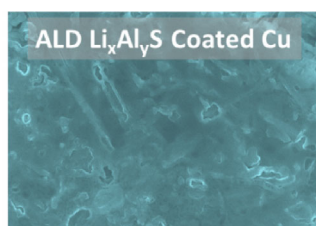
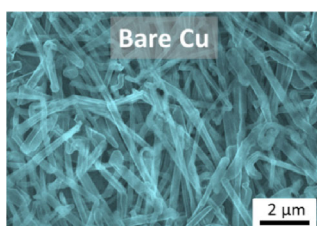
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Atomic Layer Deposition of $\text{Li}_x\text{Al}_y\text{S}$ Solid-State Electrolytes for Stabilizing Lithium-Metal Anodes

Stunted growth: Lithium metal electro-deposited onto bare copper electrode produces rough, dendritic morphology,

but atomic layer deposition of a $\text{Li}_x\text{Al}_y\text{S}$ coating with high Li^+ conductivity suppresses Li dendrite growth.



#Lithium #dendrite suppression in #electrodeposited lithium metal #anodes @argonne **SPACE RESERVED FOR IMAGE AND LINK**

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